

SUSHIL VYAS

SENIOR SOFTWARE ENGINEER (Fortran, C++)

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Profile

I have a robust background in developing CAE engineering design tools using Fortran, C++ and Python with a strong focus on numerical methods and algorithms for solving complex engineering problems, My Experience spans multiple roles where I have:

- Optimized existing codes for better performance and computational efficiency.
- Conduct thorough testing and validation of FEA / CFD codes to ensure the accuracy and reliability of simulation results.
- Alter pre-existing software systems to introduce new functionalities and improve performance using optimization method and advanced mathematics.

Experience

SENIOR SOFTWARE ENGINEER (Fortran, C++ & MFC) | Aventura Group , Bangalore, India

APRIL 2024 – PRESENT

- **Legacy System Modernization:** Modernized and optimized legacy software systems of Lloyd's Register (LR) by integrating advanced features and enhancing performance through applied mathematics and computational optimization techniques.
- **Finite Element Analysis & Structural Assessments:** Conducted finite element modeling (global/local ship structures) using Lloyd's Register and third-party tools (e.g., PATRAN), performing strength and fatigue evaluations for marine vessels.
- **Rule-Based Marine Software Development:** Engineered Lloyd's Register Shipping and marine software, aligning with classification standards (SDA, FDA3), and advanced FE analysis engines, model translation, and GUI features for structural integrity assessments.
- **Cross-Functional Collaboration:** Partnered with Business Analysts (BAs) and Subject Matter Experts (SMEs) to design and deliver intuitive, high-performance software solutions.
- **Agile Software Development Leadership:** Spearheaded software development projects using Agile methodologies, ensuring well-defined, executable, and verifiable task implementation.
- **Data-Driven SQL Development:** Results-driven SQL Developer specializing in query optimization, database performance tuning, and actionable data analytics.
- Directed full-cycle software development using **Scrum/Kanban**, translating user needs into actionable tasks with measurable outcomes.

TEAM LEAD (Fortran & C++) | Acquisy Solutions, Ahmedabad, India 2021 – 2024

- **Software Development:** Engineered CAE software for steel structure design using Fortran and C++, delivering high-performance computational tools for engineering analysis.
- **Technical Leadership:** Architected core mathematical modeling, geometric processing, and meshing modules that enhanced structural analysis capabilities and market differentiation.
- **Team Management:** Directed a 10-person Structural Engineering Development team, aligning technical roadmaps with business objectives while improving delivery timelines.
- **Product Stewardship:** Owned the full lifecycle of structural analysis products from R&D through maintenance, ensuring reliability for mission-critical engineering applications.
- **Specialized Analysis:** Developed advanced computational modules for shear stress analysis and dynamic loading problems, enabling more accurate structural integrity assessments.
- **Agile Leadership:** Implemented Scrum frameworks as Team Lead, optimizing sprint planning and retrospectives to boost team velocity while maintaining quality standards.
- **Numerical Expertise:** Applied advanced numerical methods to solve complex coupled heat and mass transfer problems in structural systems.
- **Performance Optimization:** Refactored legacy algorithms to achieve faster solution convergence while maintaining numerical precision.
- **Validation Rigor:** Established automated testing protocols for FEA/CFD codes that reduced verification time while improving result reliability.
- **System Enhancement:** Modernized legacy CAE platforms by integrating new solver technologies and user workflow improvements.
- **Development Operations:** Managed complex development pipelines using Git, JIRA and TFS, implementing CI/CD practices that reduced release cycles.

DESIGN ANALYS (Fortran, C++& Ansys) | J & M Cad Solutions, Ahmedabad 2017 – 2019

- **Modernized and optimized legacy software systems** by integrating new features and improving computational efficiency through **advanced mathematical optimization techniques**.
- **Leveraged numerical methods and high-performance algorithms** to develop robust solutions for complex engineering problems, including fluid dynamics and structural mechanics.
- **Developed custom simulation tools** using **User-Defined Functions (UDFs)** to model fluid-structure interaction (FSI) and Multiphysics systems.
- **Enhanced initial engineering designs** through **parametric optimization** and **CAD-driven refinements**, improving performance and manufacturability.
- **Performed high-fidelity simulations** with expertise in **mesh generation, model validation, and post-processing** to ensure accurate and reliable results.
- **Validated numerical models** using **Fortran and C++**, ensuring compliance with theoretical predictions and experimental data.

MECHANICAL DESIGN ENGINEER | MSBC Group, Ahmedabad, India **2016 – 2017**

- Streamlined the design development process, implementing optimizations that increased production efficiency.
- Deployed to London to lead thermal analysis of structural glazing systems, conducting advanced heat and mass transfer simulations to validate performance.
- Developed comprehensive technical documentation, including engineering specifications and manufacturing guidelines, to ensure process standardization.
- Technical Stack: Fortran (legacy systems), C++ (high-performance modules), Python (automation & data analysis).

CADD ENGINEER | G&G Infotech, Ahmedabad, India **2016 – 2016**

- Conduct research on client industry operating procedure and existing machinery specifications to facilitate the design and tools and equipment seamlessly adaptable to current production environment.
- Conduct training session to various company employees, faculty members and students in CAD, CAM and CAE software.

DESIGN ENGINEER | Thermosense Electrinstruments Pvt Ltd, Udaipur, India **2013 – 2015**

- Developed comprehensive technical reports summarizing analysis findings, providing actionable recommendations for project improvements and component optimization.
- Evaluated technical design proposals by conducting thorough reviews of specifications, manuals, and engineering data to ensure compliance and feasibility.
- Designed precision CAD models aligned with client requirements, ensuring accuracy and functionality for manufacturing and implementation.
- Proficient in industry-standard design software: AutoCAD (2D/3D drafting), SolidWorks (parametric modelling), NX CAD (advanced CAD), and ANSYS (simulation & FEA).

MECHANICAL ENGINEER | Binani Cement Ltd, Mount-Abu, India **2012 – 2013**

- Prepared and maintained technical documentation, including reports, manuals, and engineering specifications to support project execution.
- Analyzed and interpreted P&ID diagrams, technical schematics, and vibration data reports to assess system performance and identify improvement opportunities.
- Developed precision 2D engineering drawings compliant with Geometric Dimensioning & Tolerancing (GD&T) standards.
- Proficient in industry-standard design tools: AutoCAD (2D drafting), NX CAD (3D modeling), and ANSYS (simulation & FEA).

Education

MASTER OF ENGINEERING (ME) IN CAD-CAM

2019 – 2021

Lalbhai Dalapathi College of Engineering (LDCE) – Ahmedabad

BACHELOR OF TECHNOLOGY (B. TECH) IN MECHANICAL ENGINEERING 2007 – 2011

Global institute of Technology (GIT) – Jaipur

Certificates

. Successfully completed the course '**The Finite Element Method for Problems in Physics**' offered by the **University of Michigan**, earning a final score of **76.5%**.

. Certificate for the - **Introduction to Hydrodynamic Stability [NPTEL]**.

Skills & Abilities

- Scientific computing using C++, Fortran and Python.
- FEA & CFD Tool development using C++, Fortran and Python
- Auto-cad, Solid Works, NX-CAD
- Ansys Workbench (FEA), Ansys Fluent (CFD)
- MFC and Data Structure

Activities and Interests

Cooking, Volleyball, Yoga, Sequence